



LESSON 9.4

ADDING AND SUBTRACTING EXPRESSIONS WITH RATIONAL COEFFICIENTS

LESSON INTRODUCTION

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

LEARNING TARGETS

- I apply properties of operations to add and subtract linear expressions with rational coefficients.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

WORKING TOWARD

MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.

ESSENTIAL QUESTIONS

- What potential mistakes do you need to be mindful of when adding and subtracting linear expressions?
- How can you apply what you have learned about combining two linear expressions to combining three or more?

LESSON NARRATIVE

This is the final lesson on adding and subtracting linear expressions. By now, students are familiar with adding and subtracting linear expressions. This lesson culminates in students adding and subtracting three linear expressions.

COMPONENT SUMMARY

9.4.1 WARM-UP (~5 MIN.)

Combine like terms in two linear expressions with rational coefficients

9.4.3 GUIDED PRACTICE (20 MIN.)

Add and subtract three linear expressions with rational coefficients and identify which are equivalent

9.4.5 WRAP-UP (~5 MIN.)

Reflection on a mistake from today's lesson

9.4.2 COLLABORATION (10 MIN.)

Error analysis for analyzing student work on operations with expressions

9.4.4 YOUR TURN! (10 MIN.)

Combine like terms to simplify three linear expressions with rational coefficients

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- (Optional) Markers
- (Optional) Cards with the expressions for 9.4.3 Guided Practice if modifying to Secret Keeper

ADDITIONAL PREPARATION

- Determine which students will work together for 9.4.2 Collaboration.
- (Optional) Create a word wall with vocabulary from Lessons 1-4.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not distribute the -1 to all terms inside of the parentheses.
- Students may describe algebraic phrases like $5 - 3$ as “Subtract 5 from 3” or similar.

THINGS TO CONSIDER

- The problems in 9.4.3 Guided Practice to 9.4.5 Wrap-Up involve adding and subtracting three expressions (and therefore more procedural steps) with rational coefficients (a weak area for many students).
- Encourage strategies for operations with no calculator. Fluency with rational numbers in Grade 7 is an expectation.
- If time is an issue, then consider assigning 9.4.4 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider intentionally scaffolding 9.4.2 Collaboration as a Think-Pair-Share language routine. Give students the opportunity to engage independently in error analysis before pairing with a shoulder partner to discuss findings. Students may discover their own misconceptions through this additional entry point.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 9.4.2 Collaboration, students discuss and correct errors representative of common misconceptions in student work. This is an opportunity for students to address misconceptions, so encourage partner pairs to describe why each error was made. This will foster discussion around not only the error but also the misconception in understanding that led to it.

DIFFERENTIATE INSTRUCTION

9.4.2 Collaboration implements both auditory and visual entry points into discussion. Consider providing roles for students that best fit their learning style as an entry point for the activity. For example, one student might illustrate and circle the errors in the work, and one might verbally explain the errors. If students are stuck, consider prompting them to work the problem on their own to compare and contrast their own work.

9.4.1 WARM-UP: BELL WORK

TEACHER MOVES

Consider a brief review of operations with fractions if necessary, as fractional coefficients are included throughout the lesson.

DIFFERENTIATE INSTRUCTION

If only a percentage of students are still struggling with operations with fractions, consider pulling a small group during 9.4.4 Your Turn! or using On-Ramp videos as scaffolding.



9.4.1 Warm-Up: Bell Work

1. Rewrite each expression by combining like terms.

$$\begin{aligned} \text{a. } & 5.7x - (3.5x + 6) \\ &= 5.7x + -1(3.5x + 6) \\ &= 5.7x + -3.5x + -6 \\ &= 2.2x - 6 \end{aligned}$$

$$\begin{aligned} \text{b. } & \left(-\frac{5}{6}y + 5\frac{1}{2}\right) - \left(\frac{2}{3}y + 34\right) \\ &= \left(-\frac{5}{6}y + 5\frac{1}{2}\right) - \frac{2}{3}y - 34 \\ &= \left(-\frac{5}{6}y - \frac{2}{3}y\right) + 5\frac{1}{2} - 34 \\ &= -1\frac{1}{2}y - 28\frac{1}{2} \end{aligned}$$



9.4.2 Collaboration: Adding and Subtracting Linear Expressions

1. Two students made errors when rewriting the expression $(6x + 9) - (7 - x) + (8x + 11)$.

Zoe's Work	Patricia's Work
$(6x + 9) - (7 - x) + (8x + 11)$	$(6x + 9) - (7 - x) + (8x + 11)$
$(6x + 9) - (7x + 18)$	$(5x + 2) + (8x + 11)$
$-x - 9$	$13x + 13$

- a. Working with your partner, write an explanation to Zoe that describes her error.
Zoe, it looks like you did not follow the order of operations or recognize that a -1 was being distributed to $(7 - x)$. You subtracted x from $8x$ and added 7 and 11 , but you can't combine these terms until you have first distributed the negative to $(7 - x)$. Once that is done, then you can combine like terms from left to right.
- b. Write an explanation to Patricia that describes her error.
Patricia, it looks like you did not distribute -1 to both terms in the expression $(7 - x)$. You subtracted 7 from 9 , but you should have added x to $6x$ rather than subtracting it because subtracting $-x$ is the same as adding x .
- c. Simplify the expression $(6x + 9) - (7 - x) + (8x + 11)$.
 $= 6x + 9 - 7 + x + 8x + 11$
 $= 15x + 13$

9.4.2 COLLABORATION: ADDING AND SUBTRACTING LINEAR EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Consider providing an additional entry point for verbal processors by intentionally scaffolding the error analysis as a Think-Pair-Share in which students first analyze independently before sharing responses with a partner for more precise responses.

ENRICHMENT

Following question 1b, give students an opportunity to formalize their learning before applying in question 1c.

- When have you made this error before?
- What do you look for to avoid this error in the future?
- What will that look like in question 1c?



9.4.3 Guided Practice: Equivalent Expressions

1. Circle the expressions that are equivalent to $-5x - 3$ and cross out those that are not.

$$\begin{aligned} &(-4\frac{1}{8}x - 2) + (0.5x + 3) - (1\frac{3}{8}x + 4) \\ &= (-4\frac{1}{8} + 0.5 - 1\frac{3}{8})x + (-2 + 3 - 4) \\ &= -5x - 3 \end{aligned}$$

$$\begin{aligned} &-2x - (8x + 5) + (x + 2) \\ &= (-2 - 8 + 1)x + (-5 + 2) \\ &= -9x - 3 \end{aligned}$$

$$-5x - 3$$

$$\begin{aligned} &(7x + 4) - (3x + 2) + (-9x - 9) \\ &= (7 - 3 - 9)x + (4 - 2 - 9) \\ &= -5x - 7 \end{aligned}$$

$$\begin{aligned} &(-\frac{1}{2}x - 6) + (3x - 7.25) - (7.5x + 1\frac{3}{4}) \\ &= (-\frac{1}{2} + 3 - 7\frac{1}{2})x + (-6 - 7.25 - 1.75) \\ &= -5x - 3 \end{aligned}$$

9.4.3 GUIDED PRACTICE: EQUIVALENT EXPRESSIONS

DIFFERENTIATE INSTRUCTION

After working through each expression, ask students to Compare and Connect how they simplified each expression. This routine supports metacognitive and metalinguistic awareness and also supports mathematical conversation. Come together as a class to review connections students noticed and clarify any potential misconceptions.

QUESTIONING

Secret Keeper

If resources permit, students could close their books in order to avoid seeing the prompt with $-5x - 3$. You can display each of the four expressions and tell the students of a secret code they must find, the secret code being $-5x - 3$. They must determine what the secret code is and exactly how many of the expressions are the secret keepers (2). Students or pairs of students must submit their final answers with both the secret code and how many of the expressions have the code to determine the winners.

9.4.4 YOUR TURN!

QUESTIONING

For extension and error analysis practice (MA.K12.MTR.4.1), prompt students to determine what a student did wrong if they got the following answers:

- a. $-1.2y-9$ (added $+2.5y$ instead of $-2.5y$)
- b. $4\frac{3}{4}x+1.8$ (added $+9$ instead of -9)
- c. $5r-15$ (added $-2r-3r$ instead of $+3r$)



9.4.4 Your Turn!

1. Rewrite each expression by combining like terms.

$$\begin{aligned}\text{a. } (1 - 4y) - (2.5y + 7) - (-0.3y + 3) \\ &= (-4y - 2.5y + 0.3y) + (1 - 7 - 3) \\ &= -6.2y - 9\end{aligned}$$

$$\begin{aligned}\text{b. } \left(3\frac{1}{2}x - 4\right) + (x - 3.2) - \left(-\frac{1}{4}x + 9\right) \\ &= \left(3\frac{1}{2}x + x + \frac{1}{4}x\right) + (-4 - 3.2 - 9) \\ &= 4\frac{3}{4}x - 16\frac{1}{5} \text{ or } 4.75x - 16.2\end{aligned}$$

$$\begin{aligned}\text{c. } -(2r + 5) - (6 - 3r) + (10r - 4) \\ &= (-2r + 3r + 10r) + (-5 - 6 - 4) \\ &= 11r - 15\end{aligned}$$

**9.4.5 Wrap-Up: Growth Mindset**

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

My favorite mistake from today was ...

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

9.4.5 WRAP-UP: GROWTH MINDSET**DIFFERENTIATE INSTRUCTION**

For students who are unsure how to begin, consider using one of the following sentence stems:

- I learned that inverse operations can...
- I got question...wrong because I... Next time I will...